Cyclic LSI: proof and verification acceleration

Published revision e1e69bc12b185695439b345fd11808e9d7b95abb • 6 November 2026

The repository update

Commit message: small changes. Repository: https://github.com/XinyuanXie-hub/cyclic-lsi-new. The original 44 tracked files are preserved exactly in old lean code. The active proof has only Solution.lean and its required owned import CyclicLsi/Definitions.lean. Challenge.lean, assumptions, constant 2, comparator, checker binaries and CI workflow are unchanged. The separate streaming harness and experiment notes were not added to the repository.

Measured result

The new solution uses 654 / 5,060 × 100 = 12.9249011858% of the original code-bearing lines (12.92% rounded). Its median runtime is 62.02619875199889 / 158.8344384239972 × 100 = 39.0508502863% of the original (39.05% rounded). Code totals cover the complete owned Solution import closure, including all helpers; blank and comment-only lines are excluded. The percentages mean proportion remaining, not percentage reduction.

Complete verification workflow	Median (4 trials)	Observed range
Original proof, official workflow	158.834 s	153.721–175.054 s
New proof, official workflow	62.026 s	61.428–64.997 s
New proof, separate streaming harness	35.553 s	28.500–39.123 s

The published proof reduces full verification time by 60.95% (2.561× ratio of medians), with a win in every matched pair. The separate streaming workflow reduces the new proof’s median by another 42.68%; this is not the workflow shipped in the repository and must not be presented as its CI runtime.

Four balanced trials per version used the same machine and scratch path, fresh compilation of all owned modules and Challenge, fresh exports, statement/axiom comparison, and Lean, NanoDa and con-ron verification. Upstream dependency caches were warm; downloads, harness compilation, source copying and archival were excluded. Machine: Intel Xeon Platinum 8573C, four-CPU quota, Lean 4.35.0. These local measurements do not promise equivalent GitHub Actions or historical ~34-second timings. All 12 matched runs passed; a further fresh verification of the staged revision passed in 64.603 s, excluded from the medians.

How the mathematics changed

1. Scalar majorant. The old proof uses derivatives and monotonicity. The new proof uses $G(t)=2t^3+3t^2-6t+1-6t^2\log(t)$ and $D(t)=t(t-1)^2$. The elementary logarithm bound gives

$G(t) \geq -4D(t)$. Polynomial identities give $G(t^2) \geq 2t^2 G(t)$ and $D(t^2) \geq 4t^2 D(t)$. Applying these identities at $\sqrt{t}$ and inducting gives $2^n G(t) \geq -4D(t)$ for every n. Unbounded powers of 2 then rule out $G(t)<0$. This removes the bespoke calculus argument, not every calculus dependency in Mathlib.

2. Scaling and representation. The old proof normalizes a function on Fin n, proves abstract energy-family lemmas, and transports between Fin n and ZMod n. The new proof stays on ZMod n and proves the entropy majorant directly at scale $\sqrt{\text{avg}(f^2)}$. This removes the global normalized-function and energy-homogeneity transport layer; local division by the scale remains.

3. Spectral estimate. Separate odd/even Fourier block proofs are replaced by one rectangle with $p=\lfloor n/3 \rfloor$ and $h=n-p-2$. Six resonant symmetries cover the cubic terms. Fiber bounds use cyclic surplus, and sums of squares $(\tau a(i+j)-a(i)a(j))^2$ produce a uniform estimate for all $n \geq 4$.

4. Final argument. Parseval and a cubic Fourier identity connect the estimate to entropy. The proof still centers $u=f-\text{avg}(f)$, and a nonnegative quartic residual $(t-a)^3(t+3a)$ closes the inequality. Neither the theorem nor its constant is weakened.

How the implementation changed

The full owned import closure shrank from 27 modules, 5,060 code-bearing lines and 6,025 physical lines to 2 modules, 654 code-bearing lines and 783 physical lines. These totals include every owned helper. Solution.lean itself becomes longer because previously separate arguments are consolidated; entry-file length alone is misleading. Con-ron verifies 29,895 declaration records for the old proof and 18,066 for the new proof. The measured proof-only gain includes both mathematical simplification and reduced module/transport overhead; the experiment does not isolate their individual shares.

The separately tested software harness overlaps the isolated canonical Challenge build/export with fresh owned compilation and streams fresh Solution bytes into the unchanged Lean checker. Complete exports still undergo the unchanged statement/axiom comparison and external kernels. It joins independent work and propagates failures. Its total gain includes direct compilation orchestration as well as overlap. A forced con-ron failure correctly returned nonzero, emitted no overall success, and waited for the other checks. This harness remains outside the published minimal proof set.

Verification and recovery

Permitted axioms remain propext, Quot.sound and Classical.choice. The backup was checked byte-for-byte against the original Git objects and successfully rebuilt independently with every gate passing. To restore, copy old lean code into a separate directory and use its pinned toolchain and dependency manifest, or check out original commit 4b4ff9a506c0aac696119542b76276238c14cb9b. Downloads are required if dependencies are absent.

New Solution SHA-256: a645c1050bf067b7fc6daeb3c0040eb765dd83a5b647a7dc6fb951030eaeaa60. Unchanged Challenge SHA-256: 96d926772ff8f23f5e389d3d1e4f0796ed45b91b7789adb23a46fddac8c004f3.

The original historical optimized source was not recovered. This release uses the independently verified V2 reconstruction and makes no cross-machine historical equivalence claim. Raw timing receipts, source snapshots, checker logs and restoration manifests remain preserved in the experiment workspace.

Published verification and README log locator

GitHub Actions run 42150919674 for published commit e1e69bc passed the unchanged comparator and all three kernels. In its downloaded log, counting the comparator step’s Run group header as line 1, con-ron acceptance is line 39, NanoDa line 41, Lean line 43, and Your solution is okay! line 44. These correspond to full job-log lines 434, 436, 438 and 439. GitHub’s rendered line numbering can differ, and future logs can shift with warnings, cache state or tooling output. The proposed README uses lines 39–44 at the user’s request; those numbers identify this specific raw step log, not a stable universal location. The README/PDF update is a private review draft and has not been published.

Independent count check for the requested wording

Reading the published Git objects directly gives 783 physical source lines versus 6,025 originally across the complete owned proof import closure (12.9958506224% remaining). Physical lines include blank and comment lines. If blank lines alone are excluded, the counts are 692 versus 5,412. If comments and blank lines are excluded, the counts are 654 versus 5,060. Solution.lean alone has 744 physical, 664 nonblank and 643 code-bearing lines; its required Definitions file adds 39 physical, 28 nonblank and 11 code-bearing lines. None of these standard counts yields 682. A comparison of 682 with approximately 6,000 would therefore be unsupported or would mix metrics.

The matched published-workflow medians give 39.0508502863% of the original runtime, not 33.5%. Even the separately tested, unpublished streaming harness gives 22.3839143981% of the original median. A single GitHub CI run or an old historical time is not a matched comparison. No recorded release comparison supports 33.5%.